Miguel Giles



Applied AI Engineer · Software for regulated manufacturing (GxP · 21 CFR Part 11)

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Available: remote, part-time or full-time

PROFILE

Applied AI Engineer with 13+ years in industrial maintenance and automation, 9 of them in a GxP-regulated pharmaceutical plant. Creator of NexIndu, an AI-powered CMMS/EAM for regulated plants: RAG on pgvector with local embeddings, hybrid deterministic/LLM routing, LLMOps and 21 CFR Part 11 controls. Looking for remote Applied AI or regulated-manufacturing software roles.

CASE STUDY: NEXINDU

NexIndu — Pre-launch · Personal project · nexindu.com.ar

NexIndu — AI-powered CMMS/EAM for regulated plants

RAG: Semantic search on pgvector with multilingual embeddings computed locally (ONNX, 384 dimensions).
ROUTING: About 40 intents are resolved deterministically; everything else goes to the LLM (Gemini), which answers with cited sources.
PART 11: Electronic signatures with re-authentication and passkeys, an immutable hash-chained audit trail and soft deletes only.
LLMOPS: Automated prompt evaluations, a fallback model, usage logging and a per-company AI budget.

OTHER PERSONAL PROJECTS

NexKio — In use at 4 kiosks

Offline-first point of sale for kiosks: keeps selling even when the internet drops.

NexQuipu — Used by 3 families

Personal finance with AI analysis of credit-card statements.

NexPulse · pokerpulse.vercel.app

Blind timer for home poker tournaments, with P2P remote control over WebRTC.

NexPlay · nex-play-one.vercel.app

Real-time online board games for family play (Supabase Realtime).

EXPERIENCE

Independent software development — 2023 – present

In my own time: NexIndu, NexKio, NexQuipu, NexPulse and NexPlay.

Maintenance & Automation Technician · B.Braun Medical Argentina, Mar del Plata — 2017 – present

Maintenance and PLC automation of production lines in a GxP-regulated pharmaceutical plant. Diagnosis and resolution of critical failures. Continuous improvement with Lean/5S.

Industrial Refrigeration Technician · Coiron — 2013 – 2017

Led installation crews for industrial cold rooms; refrigeration, electrical and pneumatic systems.

SKILLS

AI: RAG, pgvector, ONNX embeddings, Gemini API, LLMOps, multi-agent orchestration
Development: Next.js, React, TypeScript, Supabase/PostgreSQL, Vercel, GitHub Actions
Regulated: GxP, 21 CFR Part 11, ALCOA+, auditing, electronic signatures
Industry: PLC/automation, Lean/5S, pneumatics, electrical, refrigeration

EDUCATION

Diploma in Python & Artificial Intelligence — UTN Facultad Regional Delta + Instituto Linux · 144 hours · Grade 100/100 · November 2026

Electromechanical Technician — Instituto Industrial Pablo Tavelli · 2009

LANGUAGES

Spanish (native) · English (intermediate)